

Cybersecurity Boot Camp

Surveying the Cyberspace

Lesson 1.3



Class Objectives

By the end of today's class, you will be able to:

- 1 Consider roles and career pathways within the cybersecurity space.
- 2 Explain the landscape of certifications available to security professionals.
- 3 Explore what the Security+ exam is and which InfoSec pathways benefit from the Security+ certification.



Let's recap



Recap

Which nodes of a web application are **potential targets**?

1 Users and admin

2 Web login

3 Server

4 Data



In every case, an attacker could target **every node** along the chain.



What are the **different types of attacks** that cybercriminals can use to target these nodes?



Recap

Different types of attacks:

User Attacks

- 1 Social engineering
- 2 Phishing attacks
- 3 Credential reuse
- 4 Malware attacks
- 5 Person in the middle
- 6 Packet sniffing
- 7 Stolen hardware

Web Attacks

- 1 Brute force attacks
- 2 Code injection
- 3 Faulty sessions



Recap

Different types of attacks:

Server Attacks

- 1 OS exploits
- 2 Malicious software

Database Attacks

- 1 Default credentials
- 2 Unpatched database
- 3 Lack of segregation



Recap

Potential defenses:

01

Strong password policies

02

Phishing training programs

03

Anti-keylogging software

04

Session management security

05

Encryption



Recap

Potential defenses:

06

Multi-factor authentication (MFA)

07

Regular security audits

08

Software and patch management

09

Employee security training

10

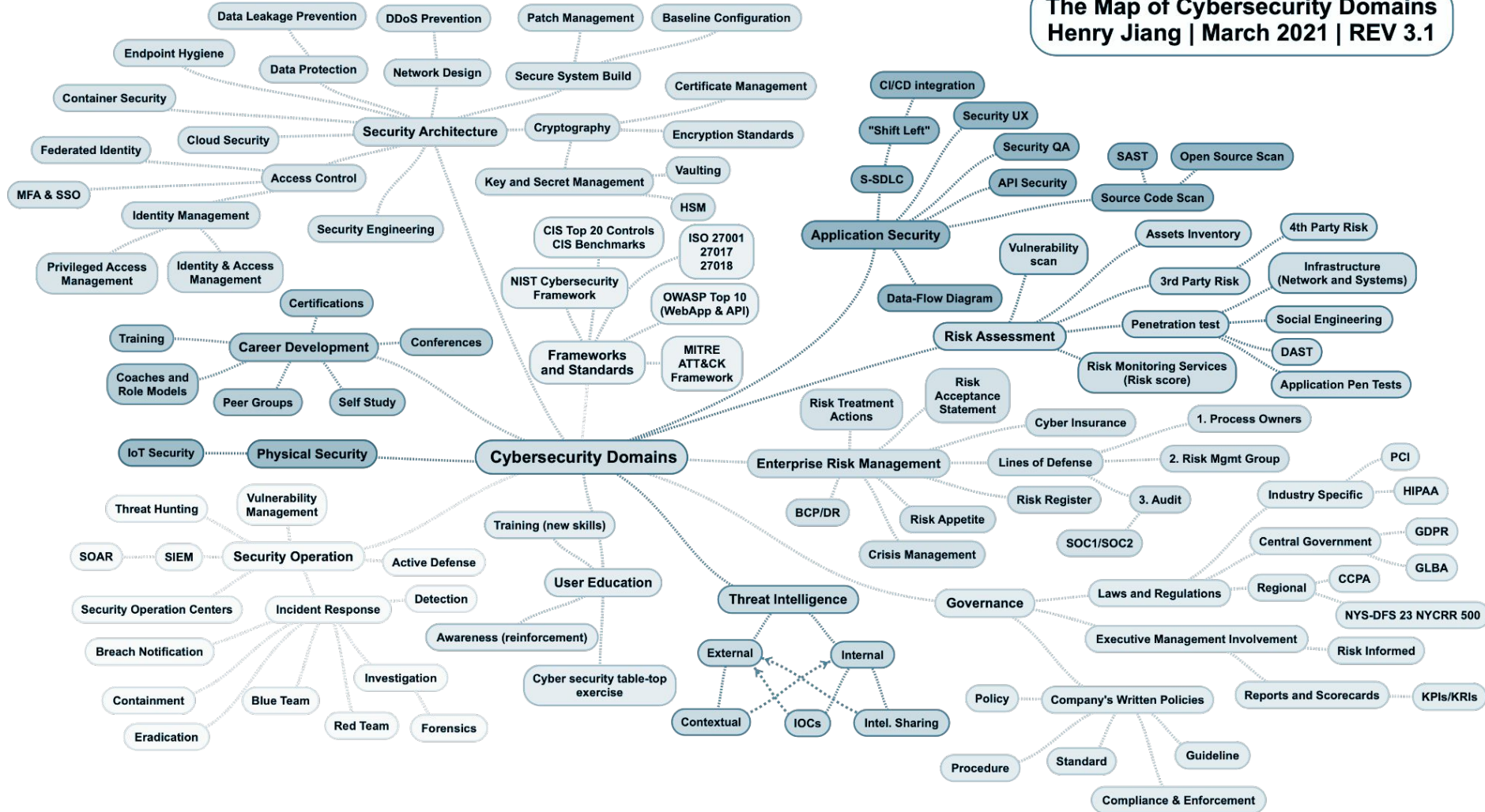
Advanced intrusion detection systems



Cybersecurity **Domains**

The Map of Cybersecurity Domains

Henry Jiang | March 2021 | REV 3.1





— Tao Security's
Richard Bejtlich
on entering the
cybersecurity field

Providing advice on getting started in digital security is similar to providing advice on getting started in medicine.

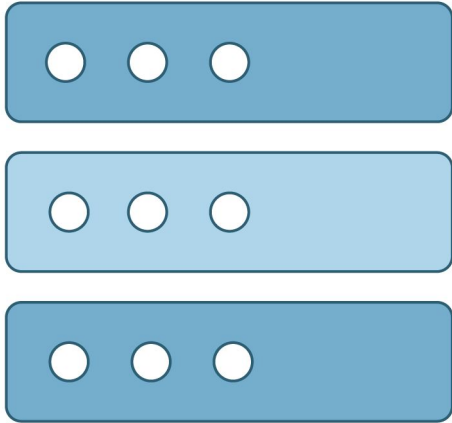
If you ask a neurosurgeon, he or she may propose some sort of experiment with dead frog legs and batteries.

If you ask a dermatologist, you might get advice on protection from the sun whenever you go outside.

Asking a 'security person' will likewise result in many different responses, depending on the individual's background and tastes.

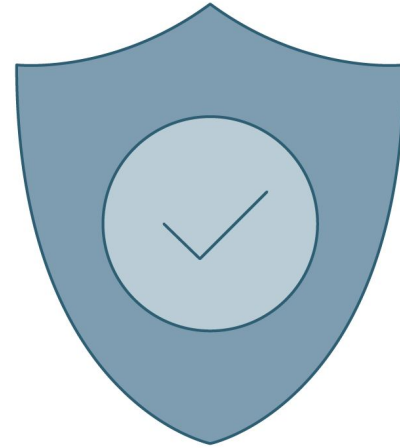
Security Architecture

Security design that addresses the necessities and potential risks



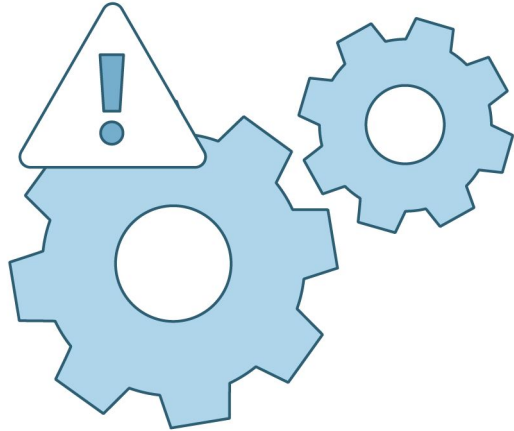
Security Operations

Identifies critical information to assess enemy observation of friendly actions and prevent adversary exploitation



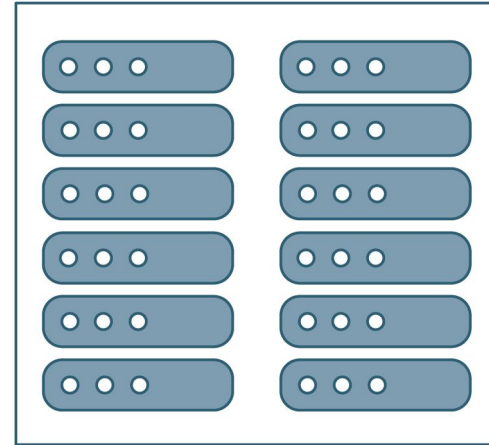
Governance

Managing performance, risk, and oversight of compliance



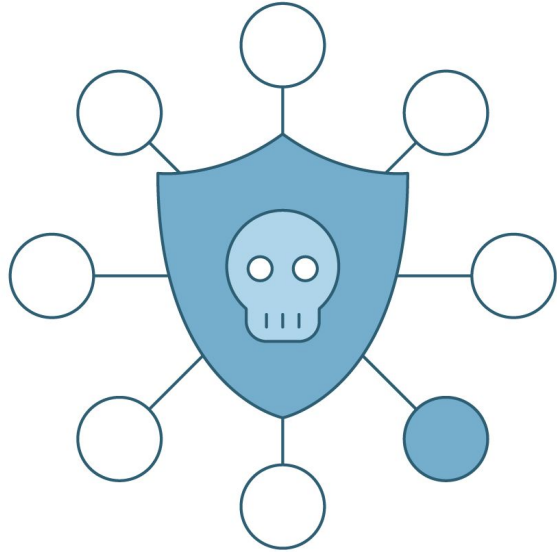
Physical Security

Protection of personnel, hardware, software, networks, and data



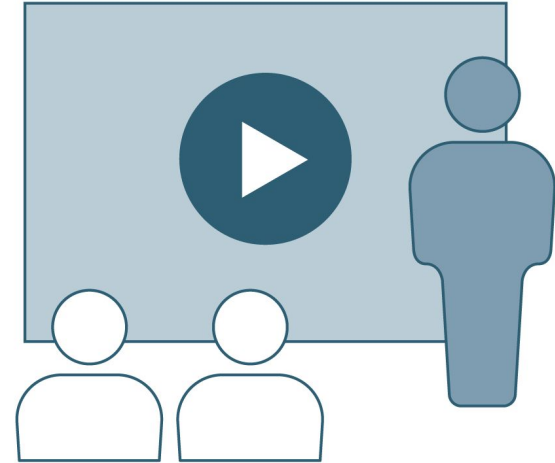
Threat Intelligence

Evidence-based knowledge about a threat to include context, mechanisms, indicators, and actionable advice



Career Development

Training and development



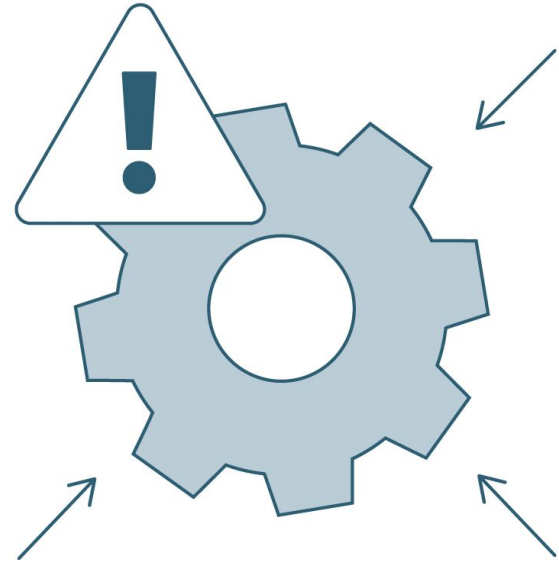
Risk Assessment

Analysis of environment and potential risks



User Education

Educating users on risks, exploits, and external threats



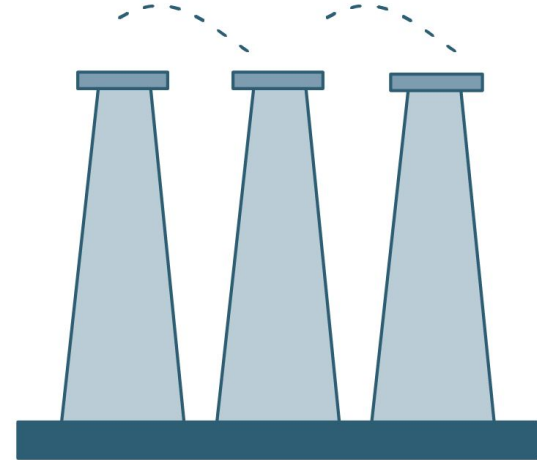
Frameworks & Standards

Creation of new security frameworks and practices



SCADA

Protecting and securing large-scale infrastructure components





Activity:

Career and Pathway Research

In this activity, we will begin to research security careers and pathways in depth.

Suggested Time:
20 Minutes





Share

Your

Answers



Employment of information security analysts is projected to grow **35%** between 2021 and 2031 — much higher than average.

— US Bureau of
Labor Statistics

Cyber Fields by the Numbers

From ISACA:

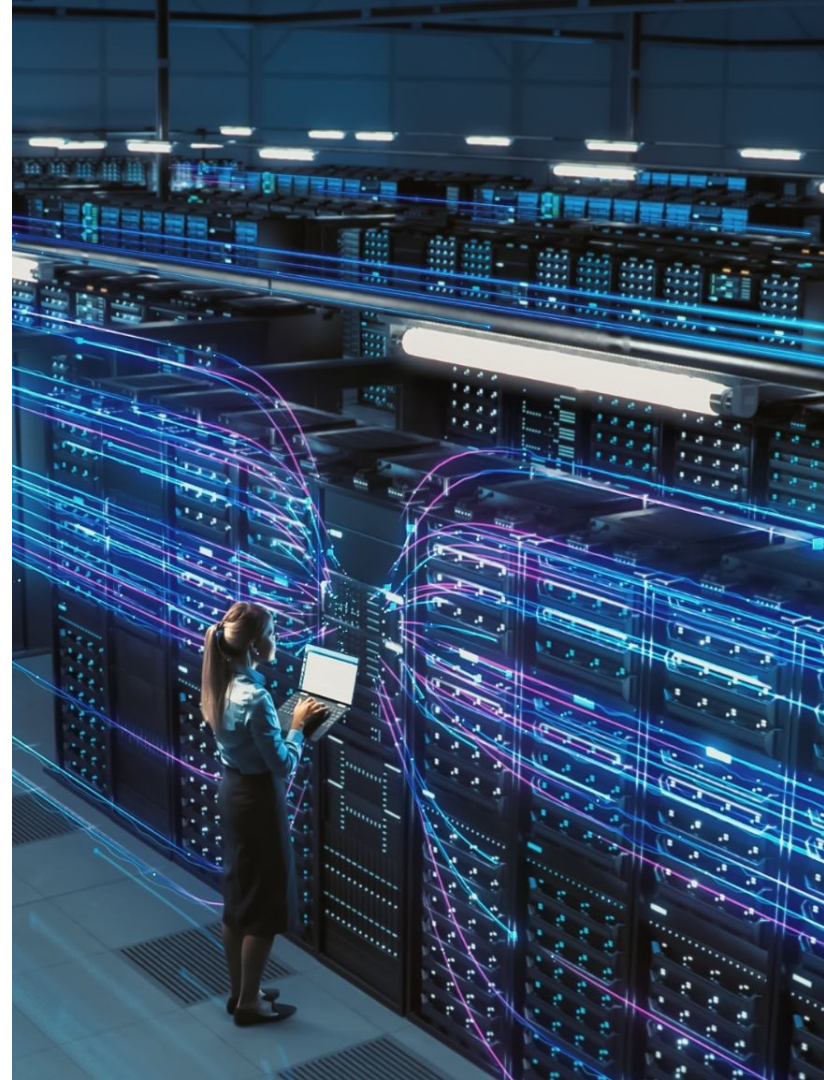
- 63% of respondent enterprises have unfulfilled cybersecurity positions.
- 60% of enterprises report experiencing difficulty retaining cybersecurity professionals.
- 82% of respondents believe that leadership sees value in cyber-risk assessments.

From (ISC)²

A full 87% of all cybersecurity professionals started their careers doing something different.

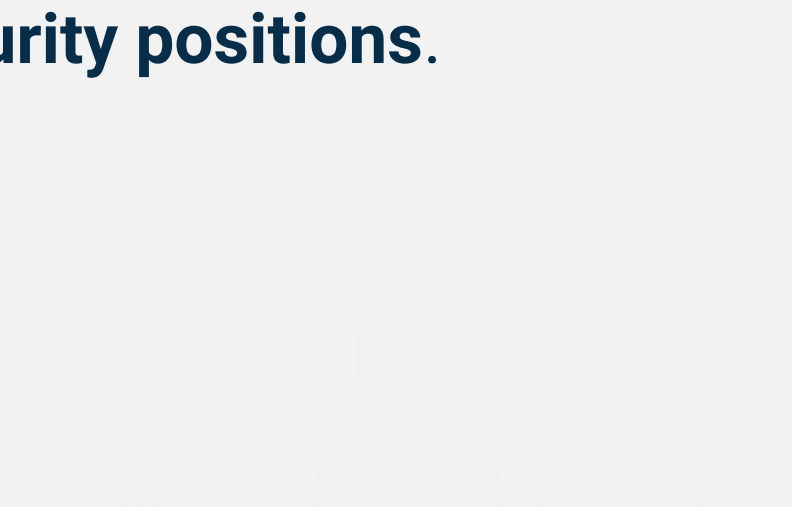
System Entry-Level Positions

- **Support/Helpdesk Specialists**
Provide technical assistance and support for users.
- **System/Network Administrators**
Manage and maintain IT infrastructure.





Let's explore some possible entry
defensive security positions.



Defensive Security Positions

- **SOC analysts**
Monitoring security systems, tier 1 SOC analysts can be hired with no prior professional experience.
- **Information specialists/analysts**
With various levels of experience, these analysts work to protect organizations from threats.





Defensive Security Positions

- **Information assurance technicians**
Protect computer assets through documentation.
- **Forensic specialists**
Analyze devices following a crime.

Defensive Security Positions



- **Risk analysts**
Examine systems and procedures to identify potential adverse events.
- **IT auditors**
Identify the weaknesses in a system's network.

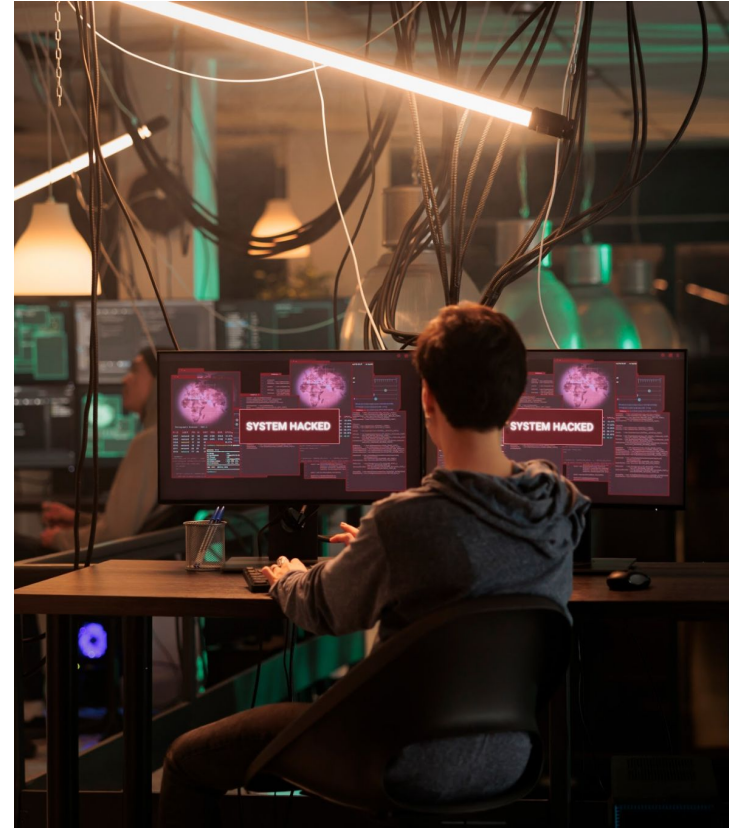


Mid-to-advanced level positions



Mid-to-Advanced Level Positions

- **Security engineers**
Test the network for vulnerabilities, monitor for security breaches, and develop security plans and policies.
- **Penetration testers/ethical hackers**
Hired by organizations to test security systems and break into security controls.



Mid-to-Advanced Level Positions

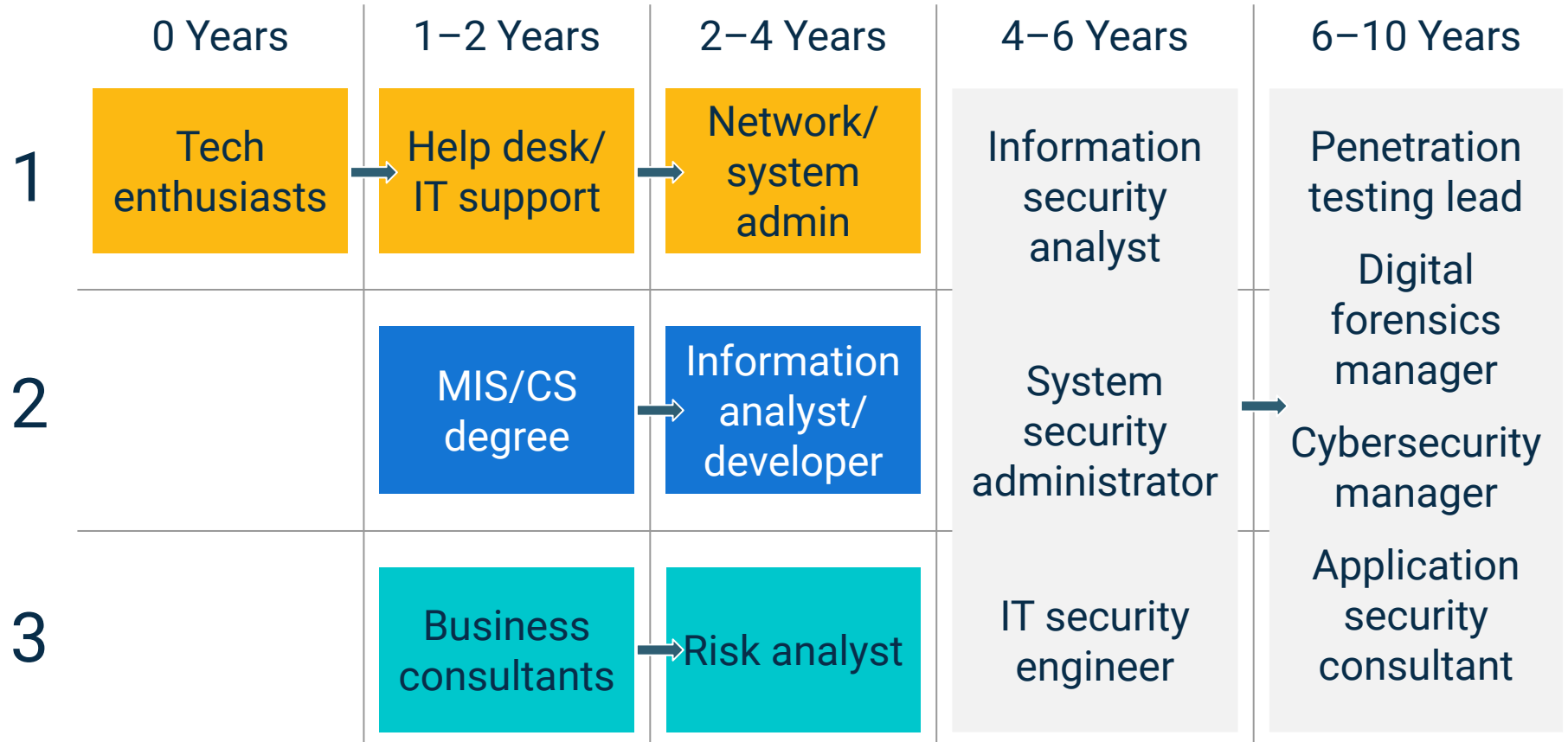
- **Chief information officer (CIO)**
Oversee and manage an organization's IT strategy, aligning with business goals.
- **Chief information security officer (CISO)**
Develop and implement an organization's information security strategy.



Expansion of Cybersecurity Positions

IoT Security Specialist	Focus on protecting networks that include smart devices in homes, factories, and cities.
AI Security Specialist	Develop strategies to protect AI systems from malicious attacks.
Cloud Security Architect	Ensure that the cloud services used by a business are secure.
Threat Hunter	Proactively search for malware or attackers that are lurking undetected.
Cyber IR Specialist	Handle security breaches and attacks.

Career Context (Common Pathways)



Career Services Department

In class:

We will tie concepts and skills in our program to career outcomes and roles.

Outside of the classroom:

The Career Services team will do the practical, hard work of helping you get a new role or promotion.

Career Services Department

There are several career milestones for you to complete in Canvas.

- You have access to them all now and can submit milestones whenever you want.
- However, you will need to submit at least **one** milestone to unlock Career Services.
- One of the milestones, an updated, polished **resume**, will be a requirement in one of our later Challenge assignments, but you can submit this earlier if you want.



Review the slides on becoming employer competitive, working with your profile coach, and working with your career director.



Break
15 mins



Security **Certifications**

Security Certifications

As the demand for cybersecurity careers grows, employers frequently use certification as a measure of employee qualifications and training when hiring.

- CompTIA certified Security+ and PenTest+.
- GIAC certifies GPEN and GCIH.
- (ISC)² certifies CISSP and SSCP.
- Offensive Security certifies OSCP and OSWP.



Of the 100+ certifications, today we'll focus on one of the most in-demand: **CompTIA Security+.**

CompTIA (Computing Technology Industry Association) is a non-profit trade organization that certifies qualified applicants in various information technology skills.

What is Security+?

According to CompTIA, Security+:

- Is the first security certification that IT professionals should earn;
- Establishes core knowledge required for any cybersecurity role;
- Provides a springboard to intermediate-level cyber jobs;
- Incorporates hands-on troubleshooting to ensure practical security problem-solving; and
- Provides testing and certification for Security+, Network+, CASP+, and PenTest+.



What jobs may require Security+?

Positions where Security+ will be helpful include:



Security architect



Security engineer



Security consultant



Security specialist



Information security analyst



As of January 2024, the
average annual pay for an
information security analyst
in the US was

\$96,652.



Security+

Domains covered in the exam:

- Attacks, threats, and vulnerabilities
- Implementation
- Architecture and design
- Operations and incident response
- Governance, risk, and compliance





Security+ is considered an entry-level exam.

The skills you'll learn in this course will offer strong foundations for many topics covered in the exam.

However, the **Sec+ exam** is broad and will require additional knowledge in areas that aren't covered in this program.

The **CompTIA CertMaster tool** will provide the material needed to close these gaps and master the exam.



This course focuses on providing relevant practical experience of the most common and useful concepts, tools, and technologies used in security and networking.

It is not a test prep course.

Security+ Exam

- 1 Some topics on the Security+ exam are not covered in this course because they are highly specific and relevant only to certain subfields of cybersecurity.
- 2 While this course and exam share many overlapping topics, more niche topics covered on the exam will require additional study.
- 3 For example, the TACACS+ protocol appears on the Security+ exam, but it is only used by engineers who work specifically with Cisco devices.

Security+ Specs

The Security+ certification is obtained by passing the CompTIA-administered Security+ exam.

There are 90 multiple choice and performance-based questions.



The exam lasts 90 minutes.



The passing score is 750 (on a scale of 100–900).



You will receive a free exam voucher at the end of the program.



Weekly challenge

1. Submit your course pre-work for the first part of this week's Challenge.
2. Write a brief summary of your current skills and what you would like to do after this program.
3. Review three prominent security reports to get a basic understanding of the market.



Fireside Chat





Thank you